

A Card-Based Design Toolkit for Co-Designing Remote Multimodal Interaction Systems with Older Adults and Experts.

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Figure 1: Sample Multimodal Social Extended Reality Cards used to scaffold co-design sessions with older adults and experts.

Abstract

This paper introduces a card-based design toolkit that enables older adults to co-design XR, spatial, and physical interaction systems. The toolkit offers a tangible, engaging, and non-intimidating way to capture their ideas and social desires, translating them into concepts that bridge complex multimodal technologies with lived experience. Embedding this approach into the design process produces interaction concepts that are both technologically innovative and socially attuned, enhancing the capacity of XR environments to support emotional well-being and long-distance social connectedness.

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1 INTRODUCTION

As society ages and family structures evolve, many older adults increasingly rely on long-distance social connections [World Health Organization 2024]. However, traditional methods such as video calls and text messaging often fall short in effectively supporting emotional expression, cognitive engagement, and sustained social participation [Burholt et al. 2023]. Recent technologies, including Extended Reality (XR)-based social platforms, avatar-mediated interactions, and wearable devices, have expanded opportunities for long-distance connections [Melo et al. 2020]. Yet limited research has explored whether multimodal interactions in XR environments can improve older adults' emotional well-being. Existing platforms and devices frequently overlook older adults' specific needs, preferences, and interaction challenges.

2 Participatory Design with Cards

2.1 Related Work

Including older adults as participants in HCI design research provide valuable insights and innovative potential for technology development. Their involvement can bring together diverse perspectives, deepen understanding of the various elements of services under development, and foster a shared understanding among people with different experiences and viewpoints [Steen et al. 2011; Tunc et al.

2023]. However, within speculative design for future technologies, collaborating with older adults from diverse backgrounds may pose a distinct set of challenges and constraints for the implementation of PD methods [Pradhan et al. 2020]. To address these challenges, card based tools commonly used in design workshops can serve as lightweight scaffolding to support PD activities and lower the entry barrier for older participants. Owing to their playfulness and combinability, cards are more likely to spark creativity, and compared with directly designing an interactive system, they can also reduce perceived intimidation [Pollmann 2021]. Cards can further explain complex concepts in accessible terms and help translate abstract ideas into concrete proposals. Accordingly, in XR contexts [Wetzel et al. 2017], HCI with older adults [Pollmann 2021], and long distance relationship communication [Li et al. 2021], card methods have been adopted and have shown promise as PD media. However, previous card-based design processes often develop content, terminology, and visuals without early input from advisors deeply engaged and familiar with the cognition and behaviour patterns of target user group, relying instead on expert feedback only in later refinement stages. This risks foundational decisions, such as language, framing, and graphics, being made without user-specific insight, reducing alignment with lived experience and interaction patterns of targeted group.

2.2 Multimodal Social Extended Reality Cards

This study adopts a heuristic structure of frameworked categorization and situated customization, drawing on the cases of PLEX Cards [Lucero and Arrasvuori 2010] and Flexi Card Game [Li et al. 2021]. Specifically, modular elements can be freely recombined to rapidly generate concepts, and new elements can be continuously added within the existing system to accommodate emerging contexts. On this basis, we developed the Multimodal Social Extended Reality Cards (Figure 1), a 49-card deck (each approximately 2.5×3.5 inch). Each card comprises four components: a reference image (generated using DALL-E 3 [OpenAI 2023]), category, keywords, and keyword explanations. The deck covers six categories: Social Partners, Interaction Mode, Physical Interaction Mode, Avatar, Scenario, and Potential Challenges.

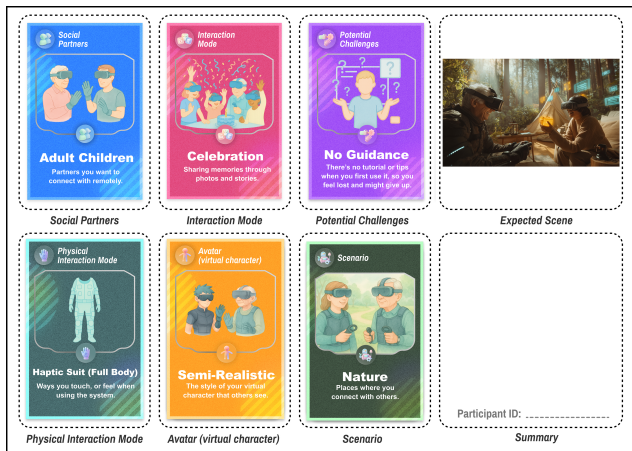


Figure 2: Cards-to-Scene Board.

The rationale for this category scheme draws on the dimensional framework for XR card design proposed by Mixed Reality Game Cards [Wetzel et al. 2017], and incorporates the opportunities and challenges of long-distance socio-technical systems for older adults discussed in [Chen and Li 2024; Zhou et al. 2025] as strategic guidance. Each category contains eight standard theme cards, plus one reusable custom card; the custom card can be used as needed during activities when existing cards do not capture participants' ideas, filling uncovered content and supporting subsequent iteration, as shown in Figure 1. For instance, sample cards include Family Connection and Friendship Circles under Social Partners, Voice Interaction and Haptic Feedback under Interaction Mode, and Mobility Limitation and Emotional Misinterpretation under Potential Challenges. These examples illustrate how the toolkit translates complex multimodal design considerations into accessible, relatable concepts for participants. In addition, we designed a Cards-to-Scene board (approximately 11.7×8.3 inch), as shown in Figure 2. Partitioned into three functional areas: (1) Card categories area, used to affix the cards selected by participants from each category; (2) Expected Scene area, which uses GenAI [Midjourney 2023] to generate and display the anticipated interaction scene; and (3) Summary area, used to record participants' revision suggestions for the scene.

3 Study Design and Future Work

PROBE	UNDERSTAND	GENERATE
INTRODUCTION The introduction will outline the workshop's aims, then have participants introduce themselves to build familiarity and a supportive atmosphere for co-design.	EXPERIENCING Participants will join a VR Christmas celebration, interacting with virtual avatars through handshakes and hugs enhanced by haptic gloves.	DOING & ADAPTING 1. Sketch-based: Write, sketch, post ideas, or use low-fidelity prototypes on zoned boards. 2. Card toolkit: Place and arrange 6 types of cards on the same boards to create interaction concepts.
		MAKING 1. Sketch-based concepts: Use participants' drawings or prototypes as image-to-image inputs in Midjourney to create the expected scenes. 2. Card-based concepts: Collect keywords from the 6 card types arranged on the boards, and use Midjourney to visualize participants' expected scenes.
		DISCUSSION 1. Compare the two co-design tools. 2. Recommend XR interaction patterns and scenarios. 3. Analyze top choices across all card categories.

Figure 3: Process overview of co-design sessions [Sanders and Stappers 2008; Tunc et al. 2023].

This study plans to adopt a Research through Design approach [Zimmerman et al. 2007], with each workshop following a process framework [Sanders and Stappers 2008; Tunc et al. 2023], as shown in Figure 3. In this study, expert consultation/involvement was embedded from the outset and continued iteratively. Advisors were researchers specialising in older adults' social interaction, with extensive, sustained engagement with ageing populations. Their input shaped early decisions on card categories, language, and graphics, and later refinements to size, usability, and visual clarity. This ensured the toolkit's conceptual framing and tangible artefacts were grounded in specialised knowledge, strengthening both contextual relevance and practical accessibility for the intended user group. In our experiment, we will employ an interactive VR Christmas celebration that is intentionally familiar and emotionally inclusive, simulating a shared family holiday to lower entry barriers and support understanding and uptake. Familiarity in virtual settings evokes sensory and embodied memory [Yang and Xiao 2024; Zhou et al. 2024]; context priming with familiar visual artifacts further

supports memory retrieval and performance, deepening engagement and presence [Hawes and Arya 2022; Zhou et al. 2025]. We will run a three-stage co-design series (2 hours per session): (1) Pilot study (n=5, internal developers) to calibrate terminology and task flow; (2) Expert workshop (n=8; XR developers, researchers, human-factors specialists, retired scholars) to assess six-category coverage and feasibility; (3) Older-adult workshop (n=8, 65+) to verify comprehensibility and contextual fit, elicit preferences and concerns, and prioritize requirements. In conclusion, we present a card-based co-design toolkit for collaboratively designing a multimodal communication systems with older adults and experts.

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